

Active Directory Domain Services (AD DS), DNS, and Group Policy Setup

Internal IT Knowledge Base / Standard Operating Procedure (SOP)

Document Type	Internal IT Knowledge Base / User Guide / SOP
Audience	IT Administrators, IT Support, System Administrators, and Non-IT Users
Environment	Windows Server / Windows 10 lab or small-organization environment
Domain	lab.local
Primary Domain Controller	DC01
Primary DNS Server	DC01 (192.168.1.100)

1. Purpose

This document provides a structured, easy-to-follow procedure for implementing and managing a basic Windows domain environment using:

- Active Directory Domain Services (AD DS)
- DNS
- Organizational Units (OUs)
- Group Policy Objects (GPOs)
- Windows workstations
- Remote Desktop Services (RDP)
- File and print server policy examples
- Basic security, auditing, password, and account-lockout policies

The goal is to provide a practical foundation similar to an internal company IT Knowledge Base / Standard Operating Procedure (SOP).

2. Audience

2.1 IT Staff

This section is primarily intended for:

- System Administrators
- IT Support
- Help Desk personnel
- Network/System engineers
- Junior IT administrators
- IT trainees

IT personnel may perform configuration, troubleshooting, user administration, GPO management, and server administration according to their assigned permissions.

2.2 Non-IT Users

Non-IT users generally do not configure AD, DNS, or GPOs.

They should understand:

- How domain login works
- Why company computers are managed centrally
- How to request access
- How Remote Desktop access is controlled
- Why certain settings such as USB access or Control Panel access may be restricted
- What information to provide when reporting an IT problem

3. Important Terminology

Term	Meaning
AD DS	Active Directory Domain Services; manages users, computers, groups, and authentication
Domain Controller (DC)	Server that hosts Active Directory and authenticates domain users/computers
DNS	Resolves names such as dc01.lab.local to IP addresses
OU	Organizational Unit used to organize users/computers and apply GPOs
GPO	Group Policy Object; centrally manages Windows settings
Domain	Logical Windows environment such as lab.local
Security Group	Group used to assign permissions or access
RDP	Remote Desktop Protocol used to remotely access a Windows computer
NLA	Network Level Authentication; requires authentication before an RDP session is established
SYSVOL	Domain directory used to store Group Policy and related domain files
DHCP	Automatically provides IP configuration to clients
NTFS	Windows file system permission model
GPResult	Tool used to determine which GPOs were applied

4. Reference Lab Environment

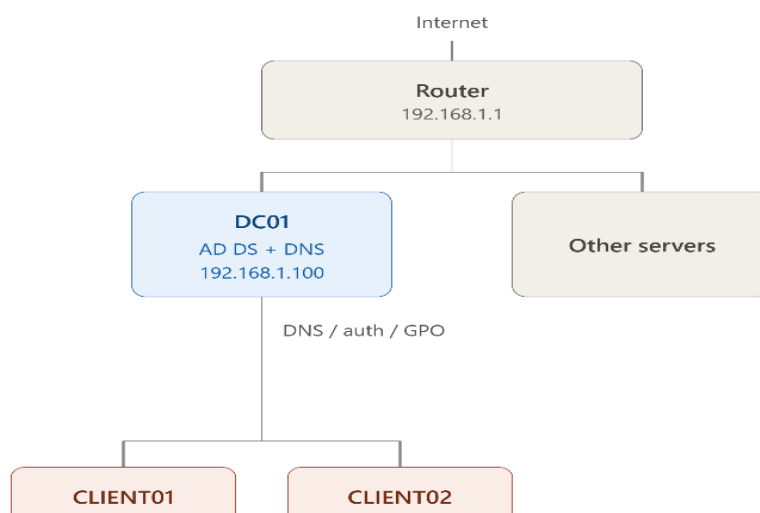
The following values are used throughout this document.

Component	Example
Domain	lab.local
NetBIOS Domain	LAB
Domain Controller	DC01
DC01 IP	192.168.1.100
DNS Server	192.168.1.100
Example Gateway	192.168.1.1

Note: The gateway and client IP addresses are examples. Use the actual addresses assigned to your environment.

5. High-Level Architecture

A basic environment can be represented as:



6. Roles and Responsibilities

6.1 IT Administrator

Responsible for:

- Installing and maintaining AD DS
- Managing DNS
- Creating and maintaining OUs
- Creating and modifying GPOs
- Managing security groups
- Managing domain users and computers
- Troubleshooting authentication and policy problems
- Reviewing security logs
- Managing RDP access

6.2 IT Support

Responsible for:

- Basic domain troubleshooting
- Checking client IP/DNS configuration
- Running gpupdate /force
- Running gpresult /r
- Checking user/group membership
- Confirming whether a GPO is applied
- Escalating domain-level issues to an administrator

6.3 End Users

Users should:

- Use their assigned domain account
- Protect their credentials
- Avoid sharing passwords
- Report authentication, network, or application problems to IT
- Follow company security policies

7. Prerequisites

Before deploying AD DS:

- Windows Server is installed
- The server has a static IP address
- The server has a unique hostname
- The server can communicate with clients
- The administrator account is available
- The planned domain name is known
- Network connectivity has been tested

For this lab:

Server:	DC01
IP:	192.168.1.100
Domain:	lab.local

8. Configure DC01 Static IP

On DC01:

```
Control Panel
→ Network and Internet
→ Network Connections
```

Right-click the network adapter:

```
Properties
→ Internet Protocol Version 4 (TCP/IPv4)
→ Properties
```

Configure:

```
IP address:      192.168.1.100
Subnet mask:     255.255.255.0
Default gateway: 192.168.1.1
Preferred DNS:   127.0.0.1
```

Verify:

```
ipconfig /all
```

The server should show:

```
IPv4 Address:      192.168.1.100
DNS Server:        127.0.0.1
```

9. Rename the Server

Open:

```
Server Manager
→ Local Server
```

Change the computer name to:

```
DC01
```

Restart the server.

Verify:

```
hostname
```

Expected:

```
DC01
```

10. Install Active Directory Domain Services

Open:

```
Server Manager
→ Manage
→ Add Roles and Features
```

Select:

```
Role-based or feature-based installation
```

Select:

```
DC01
```

Select:

```
Active Directory Domain Services
```

When prompted:

```
Add Features
```

Continue and select:

```
Install
```

11. Promote DC01 to a Domain Controller

After AD DS installation:

Server Manager
→ Notification Flag
→ Promote this server to a domain controller

Select:

Add a new forest

Root domain:

lab.local

12. Domain Controller Options

Configure:

DNS Server:	Enabled
Global Catalog:	Enabled
Read-only Domain Controller:	Disabled

Create a Directory Services Restore Mode (DSRM) password.

Continue through the wizard.

13. DNS Delegation

A warning may appear stating that a DNS delegation cannot be created.

For this standalone lab environment, continue through the wizard.

14. NetBIOS Domain Name

The expected NetBIOS name is:

LAB

Therefore:

DNS domain:	lab.local
NetBIOS domain:	LAB

15. Database and SYSVOL Paths

For a basic lab, keep the default paths:

C:\Windows\NTDS
C:\Windows\SYSVOL

Complete the installation.

The server will restart.

16. Domain Administrator Login

After the restart, use:

LAB\Administrator

or:

Administrator@lab.local

17. Verify AD DS

Run:

```
Get-Service NTDS
```

Expected:

Status	Name
-----	----
Running	NTDS

Then:

```
Get-ADDomain
```

Confirm that:

DNSRoot	:	lab.local
NetBIOSName	:	LAB

18. Verify DNS

Run:

```
Get-Service DNS
```

Expected:

Status	Name
-----	----
Running	DNS

Test:

```
nslookup lab.local
```

and:

```
nslookup dc01.lab.local
```

The domain controller should resolve to:

```
192.168.1.100
```

19. Client DNS Configuration

This is one of the most important AD requirements.

CLIENT01 should use the AD DNS server:

```
192.168.1.100
```

Example:

IP address:	192.168.1.120
Subnet mask:	255.255.255.0
Default gateway:	192.168.1.1
Preferred DNS:	192.168.1.100

Note: Do not use public DNS as the client's primary DNS for domain discovery:

```
8.8.8.8
1.1.1.1
```

Verify:

```
ipconfig /all
```

Then:

```
nslookup lab.local
nslookup dc01.lab.local
```

20. Test AD DNS Service Records

On CLIENT01:

```
nslookup
```

Then:

```
set type=SRV
_ldap._tcp.dc._msdcs.lab.local
```

The result should identify DC01.

If the client cannot resolve the AD SRV records, troubleshoot DNS before attempting to join the domain.

21. Join CLIENT01 to the Domain

On CLIENT01:

```
Settings
→ System
→ About
→ Advanced system settings
```

or:

```
Control Panel
→ System
→ Advanced system settings
→ Computer Name
→ Change
```

Select:

```
Domain
```

Enter:

```
lab.local
```

Provide authorized domain credentials.

Example:

```
LAB\Administrator
```

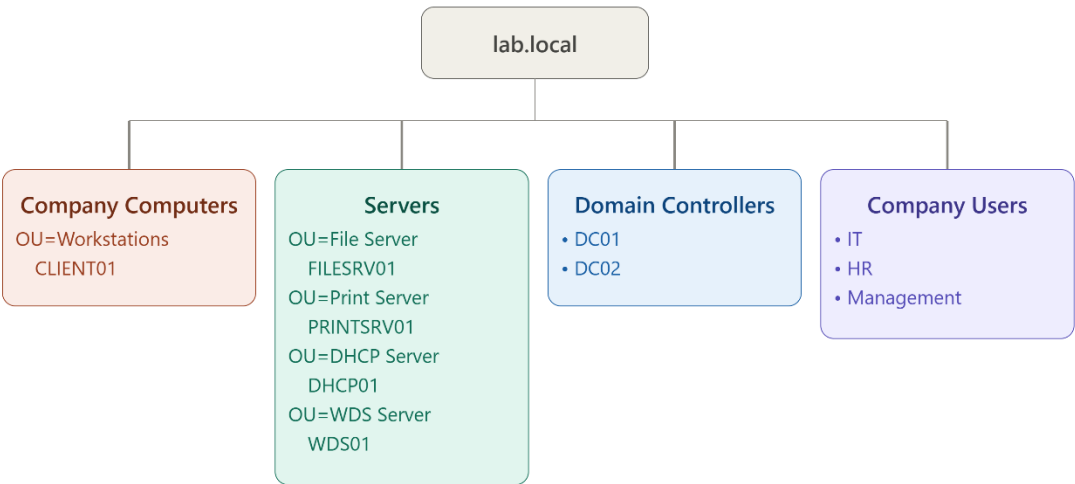
After successful joining, Windows should display:

```
Welcome to the lab.local domain.
```

Restart the computer.

22. Active Directory OU Structure

Create a structured OU design. The recommended layout for this lab is:



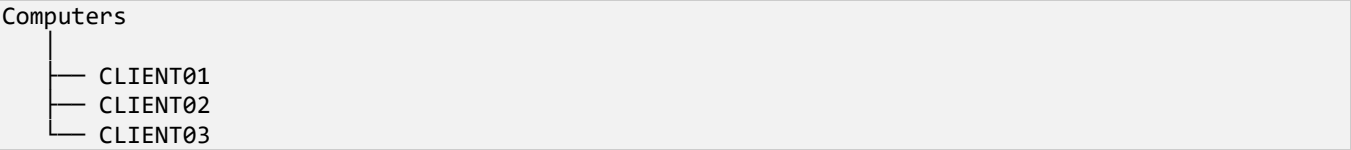
Note: You don't normally move DC01/DC02 out of the built-in Domain Controllers OU.

The OU structure determines where users and computers are managed and where GPOs are linked.

22.1 What Is an OU?

An Organizational Unit (OU) is basically a container that lets you organize computers and users.

For example:



You can then say:

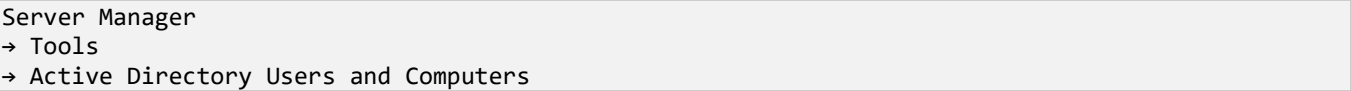
Principle: “Apply this policy to everything inside Computers.”

That's much better than configuring every computer individually.

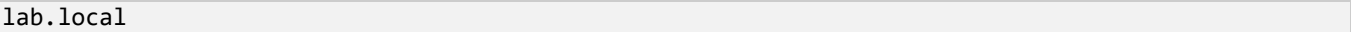
22.2 Create Your OUs

On DC01:

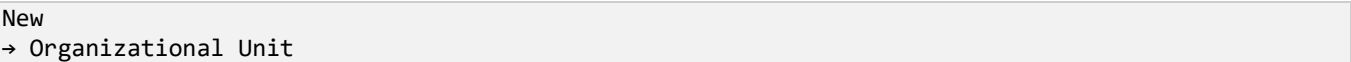
Open:



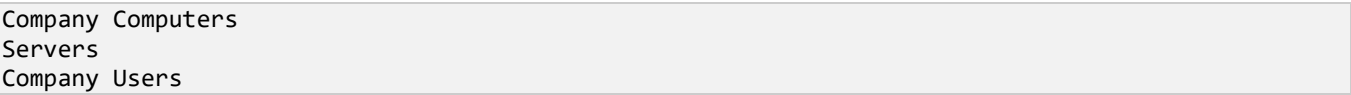
Right-click your domain:



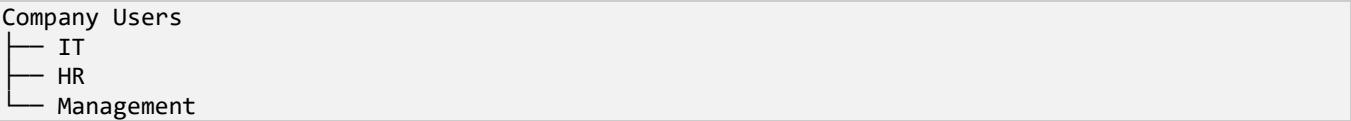
Choose:



Create:

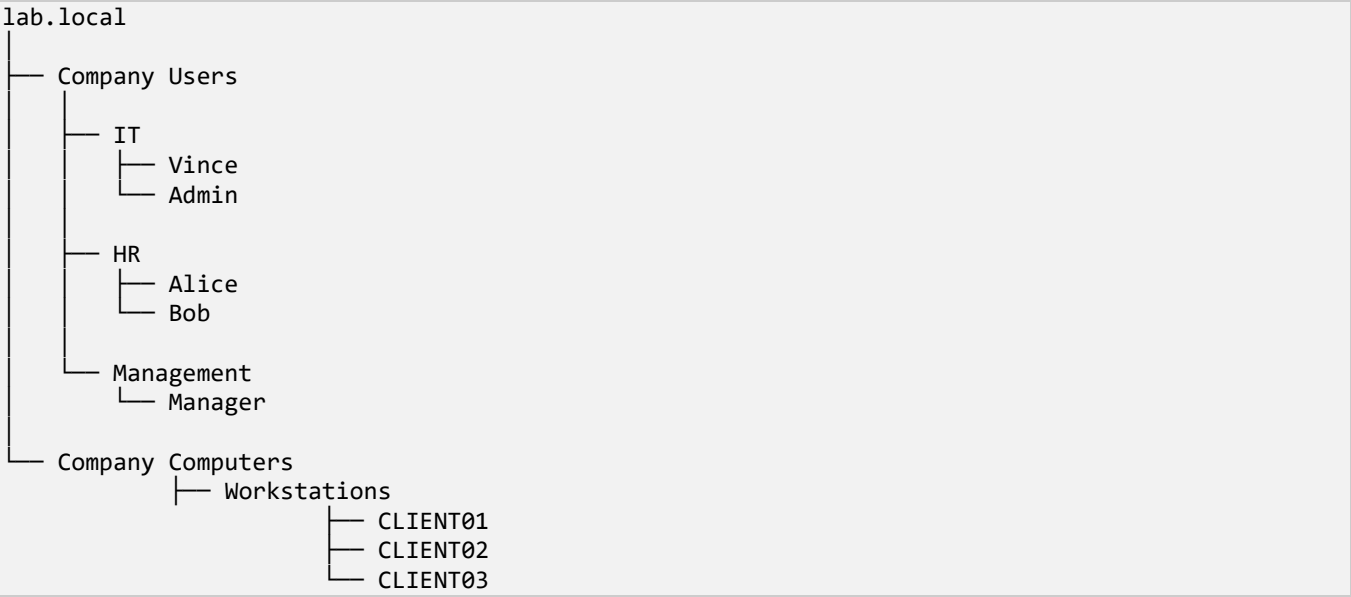


Then create the required sub-OUs inside each parent OU, for example:



Repeat the same New → Organizational Unit steps inside Company Computers and Servers to create Workstations, File Server, Print Server, DHCP Server, and WDS Server.

Once populated with users and computers, the tree looks like this:



23. Create Active Directory Users

Once the OU structure exists, create the domain user accounts inside the appropriate OU.

23.1 Open Active Directory Users and Computers

On DC01, open:

```
Server Manager
→ Tools
→ Active Directory Users and Computers
```

Expand:

```
lab.local
├── Company Users
│   └── IT
```

23.2 Create a New User

Right-click the target OU, for example:

```
IT
```

Choose:

```
New
→ User
```

Fill in the name fields.

Example:

```
First name:      Vince
Full name:       Vince
User logon name: vince
```

Click:

```
Next
```

23.3 Set the Password

Set an initial password.

Recommended options for a new account:

```
Password: (temporary/initial password)
☒ User must change password at next logon
☐ User cannot change password
☐ Password never expires
☐ Account is disabled
```

Note: Do not enable “Password never expires” for normal user accounts. It is normally reserved for dedicated service accounts.

Click:

```
Next
→ Finish
```

23.4 Verify the User

The new account should now appear under:

```
Company Users
├── IT
│   └── Vince
```

Confirm from PowerShell:

```
Get-ADUser -Identity vince
```

Expected:

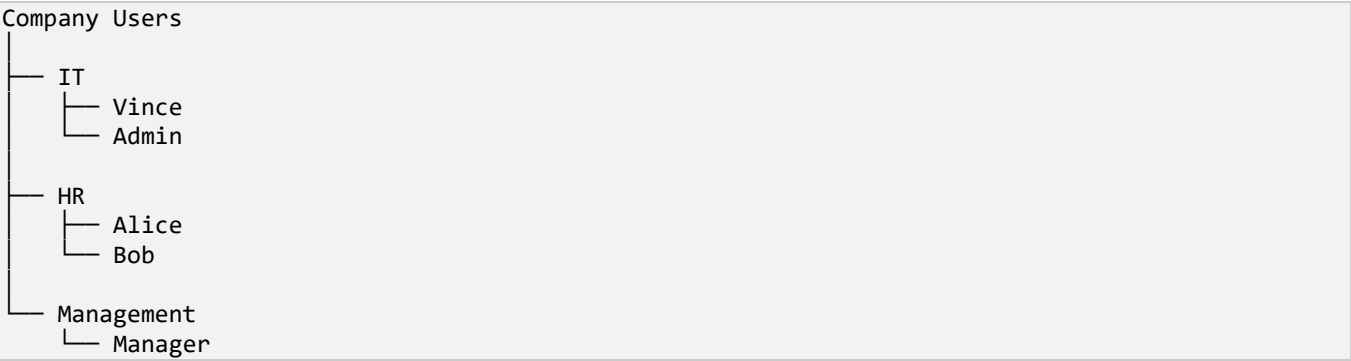
```
Name           : Vince
Enabled         : True
DistinguishedName : CN=Vince,OU=IT,OU=Company Users,DC=lab,DC=local
```

23.5 Example Users for This Lab

Repeat Steps 23.2–23.4 to populate every OU. The recommended set of example accounts for testing is:

User	OU	Purpose
Vince	IT	IT staff / lab administrator account
Admin	IT	Secondary IT/admin test account
Alice	HR	HR staff test account
Bob	HR	HR staff test account
Manager	Management	Management test account

Resulting structure:



Note: Being a member of the IT OU does not automatically make a user a Domain Admin. Administrative rights are granted separately through group membership, not OU placement.

23.6 Add a User to a Security Group

Some GPOs and permissions (such as GPO - Remote Desktop in Section 36) rely on security-group membership rather than OU placement.

Right-click the user:

Vince

→ Properties

→ Member Of

→ Add...

Type the group name:

GG-RemoteDesktop-Users

Click:

Check Names

→ OK

Confirm from PowerShell:

Get-ADGroupMember -Identity GG-RemoteDesktop-Users

23.7 Bulk-Verify Users With PowerShell

To list every user in an OU:

Get-ADUser -SearchBase "OU=IT,OU=Company Users,DC=lab,DC=local" -Filter *

To list every user in the domain:

Get-ADUser -Filter * | Select-Object Name, DistinguishedName

24. GPO Design Principles

Do not create one large:

GPO - EVERYTHING

Instead, separate policies by purpose.

Recommended GPOs:

GPO - Workstation Security
GPO - Workstation Restrictions
GPO - User Security
GPO - Company Branding
GPO - Network Drives
GPO - Removable Storage
GPO - Windows Update
GPO - Security Auditing
GPO - IT Users
GPO - HR Users
GPO - Management Users
GPO - Remote Desktop
GPO - File Server Security
GPO - Print Server Security
GPO - Domain Controller Security

Separate policies make troubleshooting easier.

25. GPO - Workstation Security

Target:

Company Computers
└─ Workstations

Go to:

Computer Configuration
→ Policies
→ Windows Settings
→ Security Settings
→ Windows Defender Firewall with Advanced Security

Recommended baseline:

Domain Profile
Firewall: ON
Inbound: Block
Outbound: Allow

Private Profile
Firewall: ON
Inbound: Block
Outbound: Allow

Public Profile
Firewall: ON
Inbound: Block
Outbound: Allow

The workstation must still communicate with:

DC01
DNS
DHCP
FILESRV01
PRINT01

Note: Do not block all outbound traffic.

26. GPO - Workstation Restrictions

Target:

Company Computers

- Workstations

Go to:

User Configuration

- Policies
- Administrative Templates

Control Panel

Control Panel

- Prohibit access to Control Panel and PC settings

Set:

Enabled

Command Prompt

System

- Prevent access to the command prompt

For the learning environment:

Not Configured

This keeps troubleshooting tools available.

Registry Editor

System

- Prevent access to registry editing tools

For a standard-user exercise:

Enabled

27. GPO - User Security

Target:

Company Users

Configure:

User Configuration

- Policies
- Administrative Templates
- Control Panel
- Personalization

Recommended:

Enable screen saver

- Enabled

Password protect the screen saver

- Enabled

Screen saver timeout

- Enabled
- 600 seconds

Result:

10 minutes inactive

- ↓

Screen saver

- ↓

Password required

28. GPO - Company Branding

Target:

Company Users

Go to:

User Configuration
→ Policies
→ Administrative Templates
→ Desktop
→ Desktop

Configure:

Desktop Wallpaper
→ Enabled

Example path:

\\FILESRV01\CompanyData\wallpaper.jpg

Wallpaper style:

Fill

The user computer must have permission to access the file.

29. GPO - Network Drives

Use Group Policy Preferences:

User Configuration
→ Preferences
→ Windows Settings
→ Drive Maps

Create:

Drive Maps
→ New
→ Mapped Drive

Example:

Action: Update
Location: \\FILESRV01\CompanyData
Drive Letter: Z:

Department examples:

IT
Z: → \\FILESRV01\IT

HR
Z: → \\FILESRV01\HR

Management
Z: → \\FILESRV01\Management

30. GPO - Removable Storage

Target:

Company Computers
└─ Workstations

Go to:

Computer Configuration
→ Policies
→ Administrative Templates
→ System
→ Removable Storage Access

Configure:

All Removable Storage classes: Deny all access
→ Enabled

Use this only on the intended workstation OU.

Note: Do not apply the workstation USB restriction to the host computer or servers unless that is specifically required.

31. GPO - Windows Update

Target:

Company Computers
└─ Workstations

Go to:

Computer Configuration
→ Policies
→ Administrative Templates
→ Windows Components
→ Windows Update

For the basic lab:

Configure Automatic Updates
→ Not Configured

Note: Do not disable Windows Update merely to create a policy.

32. GPO - Security Auditing

Go to:

Computer Configuration
→ Policies
→ Windows Settings
→ Security Settings
→ Advanced Audit Policy Configuration
→ System Audit Policies

Recommended practice settings:

Account Logon
Audit Credential Validation
→ Success
→ Failure

Account Management
Audit User Account Management
→ Success
→ Failure

Logon/Logoff
Audit Logon
→ Success
→ Failure

Audit Logoff
→ Success

Policy Change
Audit Audit Policy Change
→ Success
→ Failure

System
Audit Security System Extension
→ Success

Note: Do not enable every auditing category without a reason because excessive auditing can create unnecessary event volume.

33. GPO - IT Users

Target:

Company Users

└ IT

Keep troubleshooting tools available:

Command Prompt → Not Configured

PowerShell → Not Configured

Registry Editor → Not Configured

MMC → Not Configured

Control Panel → Not Configured

Use the GPO for:

- IT wallpaper
- IT network drive
- IT shortcuts

Being a member of the IT OU does not automatically make a user a Domain Admin.

34. GPO - HR Users

Target:

Company Users

└ HR

Example settings:

Control Panel restriction

→ Enabled

Registry Editor restriction

→ Enabled

Command Prompt

→ Not Configured

Also configure:

- Company wallpaper
- HR network drive

35. GPO - Management Users

Target:

Company Users

└ Management

Recommended:

Screen saver

→ Enabled

Password protect screen saver

→ Enabled

Company wallpaper

→ Enabled

Network drive

→ Enabled

Note: Do not grant administrative rights through this GPO.

36. GPO - Remote Desktop

The purpose of this GPO is to centrally enable and control Remote Desktop on workstation computers.

Target:

Company Computers

- Workstations

Create:

GPO - Remote Desktop

35.1 Enable Remote Desktop

Edit:

Computer Configuration

- Policies
- Administrative Templates
- Windows Components
- Remote Desktop Services
- Remote Desktop Session Host
- Connections

Find:

Allow users to connect remotely by using Remote Desktop Services

Set:

Enabled

35.2 Require Network Level Authentication

Go to:

Remote Desktop Session Host

- Security

Find:

Require user authentication for remote connections by using Network Level Authentication

Set:

Enabled

This is the recommended setting for the lab.

35.3 Security Layer

Find:

Require use of specific security layer for remote (RDP) connections

For a modern Windows lab:

Not Configured

Allow Windows to negotiate an appropriate security layer unless you are specifically testing RDP security-layer configuration.

35.4 Create a Dedicated RDP Security Group

On DC01:

Active Directory Users and Computers

Create:

Company Users

- GG-RemoteDesktop-Users

Recommended:

Group scope: Global
Group type: Security

Add authorized users to the group.

Example:

```
GG-RemoteDesktop-Users
└─ Vince
```

35.5 Add the Group to Remote Desktop Users

A scalable approach is to manage the local group through Group Policy Preferences.

Edit:

```
GPO - Remote Desktop
```

Go to:

```
Computer Configuration
→ Preferences
→ Control Panel Settings
→ Local Users and Groups
```

Create:

```
Local Users and Groups
→ New
→ Local Group
```

Configure:

```
Action: Update
Group name: Remote Desktop Users
```

Add:

```
LAB\GG-RemoteDesktop-Users
```

This centrally manages who is allowed to use RDP.

35.6 Enable the RDP Firewall Rule

Go to:

```
Computer Configuration
→ Policies
→ Windows Settings
→ Security Settings
→ Windows Defender Firewall with Advanced Security
→ Inbound Rules
```

Enable the built-in RDP rule similar to:

```
Remote Desktop - User Mode (TCP-In)
```

RDP normally uses:

```
TCP 3389
```

You generally do not need to create a custom rule if the built-in RDP firewall rule is available.

35.7 Apply the GPO

On CLIENT01:

```
gpupdate /force
```

Restart:

```
shutdown /r /t 0
```

After reboot:

```
gpresult /r
```

Confirm:

```
GPO - Remote Desktop
```

appears under applied computer policies.

35.8 Verify RDP

On CLIENT01:

```
Get-ItemProperty "HKLM:\System\CurrentControlSet\Control\Terminal Server" -Name fDenyTSConnections
```

Expected:

```
fDenyTSConnections : 0
```

0 means RDP connections are allowed.

Check the local RDP group:

```
net localgroup "Remote Desktop Users"
```

Confirm that the appropriate domain group is present.

35.9 Connect to CLIENT01

Find the workstation IP:

```
ipconfig
```

Example:

```
IPv4 Address: 192.168.1.120
```

From another Windows computer:

```
Win + R  
→ mstsc
```

Enter:

```
192.168.1.120
```

or:

```
CLIENT01
```

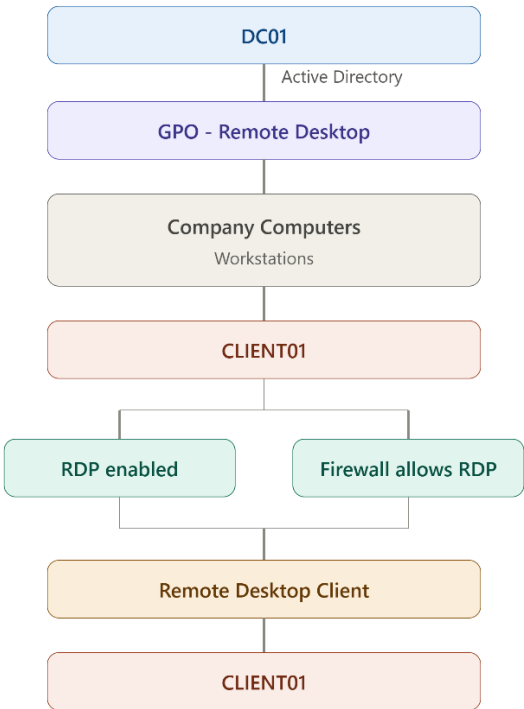
Use an authorized account:

```
LAB\Vince
```

or:

```
vince@lab.local
```

35.10 RDP Architecture



You do not need a separate RDP server VM for this scenario.

CLIENT01 itself becomes the RDP host when Remote Desktop is enabled.

37. GPO - File Server Security

Target:

Servers

- File Servers

For FILESRV01.

Configure:

Firewall → ON
Inbound → Block
Outbound → Allow

Auditing:

Audit Logon
→ Success + Failure

Audit User Account Management
→ Success + Failure

File access auditing should only be configured when required and should be combined with appropriate NTFS auditing on the target folders.

38. FILESRV01 — Active Directory Groups & Share Permissions

GPO - File Server Security (Section 37) locks down the firewall and auditing on FILESRV01, but it does not control who can read or write to each department's files. That is handled separately, through security groups and NTFS/share permissions.

This section covers creating the department security groups and applying them to the folder structure on FILESRV01.

38.1 Principle: Groups, Not Individual Users

Note: Never assign NTFS or share permissions directly to an individual user. Assign permissions to a security group, then add or remove users from that group.

This keeps permissions manageable as staff join, move between departments, or leave the organization — you only ever edit group membership, never the folder permissions themselves.

38.2 Recommended Group Design

On DC01, the Company Users OU already contains the department OUs:

Company Users

- IT
- HR
- Management

Create one global security group per department, plus a Finance group for this example, so every department that has a matching share also has a matching group:

GG_IT
GG_HR
GG_Management

Note: The GG_ prefix is a naming convention for a Global Group. Keeping a consistent prefix (GG_ for department groups, GG-RemoteDesktop-Users for access groups, etc.) makes groups easy to identify in Active Directory Users and Computers and in share permission dialogs.

38.3 Create the Security Groups

On DC01, open:

Server Manager
→ Tools
→ Active Directory Users and Computers

Right-click:

Company Users

Choose:

New
→ Group

Configure:

Group name: GG_IT
Group scope: Global
Group type: Security

Click:

OK

Repeat for each remaining group:

GG_HR
GG_Management

Confirm from PowerShell:

```
Get-ADGroup -Filter "Name -like 'GG_*'" | Select-Object Name, GroupScope, GroupCategory
```

38.4 Add Users to the Matching Group

Add each user to the group for their department. Right-click the user:

Alice
→ Properties
→ Member Of
→ Add...

Type the group name:

GG_HR

Click:

Check Names
→ OK

Or add multiple members at once from PowerShell:

```
Add-ADGroupMember -Identity GG_IT -Members vince,admin  
Add-ADGroupMember -Identity GG_HR -Members alice,bob  
Add-ADGroupMember -Identity GG_Management -Members manager
```

Recommended mapping for this lab:

Users	OU	Group
Vince, Admin	IT	GG_IT
Alice, Bob	HR	GG_HR
Manager	Management	GG_Management

Note: Domain Users is a built-in group that already contains every domain account. Do not add users to it manually — it is used below only for the shared Public folder.

38.5 Create the Folder Structure on FILESRV01

On FILESRV01, create the top-level company data folder and one subfolder per department.

Example:

```
D:\CompanyData
├── Public
├── IT
├── HR
└── Management
```

Keep company data on a dedicated data volume (D:) rather than the operating system volume where possible.

38.6 Recommended Permission Model

Apply the following permissions once the folders and groups both exist:

Folder	Group	Permission
Public	Domain Users	Read/Write
IT	GG_IT	Modify
HR	GG_HR	Modify
Management	GG_Management	Modify

This is much closer to how an actual company environment works: everyone can collaborate in Public, but each department's folder is only accessible to its own group.

Note: Modify allows a group to read, write, and delete files within the folder, but not to change NTFS permissions on it. Reserve Full Control for IT/administrative accounts only.

38.7 Apply NTFS Permissions

On FILESRV01, right-click the target folder:

```
IT
→ Properties
→ Security
→ Edit...
```

Click:

```
Add...
```

Type the group name:

```
GG_IT
```

Click:

```
Check Names
→ OK
```

Select the group, then check:

```
Modify
Read & Execute
List Folder Contents
Read
Write
```

Click:

```
Apply
→ OK
```

Repeat for HR, Finance, and Management using their matching groups, and for Public using Domain Users.

38.8 Create and Share the Folders

Right-click the top-level folder:

```
CompanyData
→ Properties
→ Sharing
→ Advanced Sharing
```

Check:

```
Share this folder
```

Share name:

```
CompanyData
```

Click:

```
Permissions
```

For the share-level permissions, grant Full Control to Authenticated Users and rely on the NTFS permissions from Step 38.7 to enforce the actual department restrictions:

```
Authenticated Users → Full Control (share)
```

Note: Share permissions and NTFS permissions are evaluated together, and the more restrictive of the two always wins. Keeping the share permission open and doing the real restriction at the NTFS level (Section 38.7) is the standard, easier-to-manage approach.

The resulting UNC paths are:

```
\\FILESRV01\CompanyData\Public
\\FILESRV01\Departments\IT
\\FILESRV01\Departments\HR
\\FILESRV01\Departments\Finance
\\FILESRV01\Departments\Management
```

38.9 Connect Permissions Back to GPO - Network Drives

Section 28 (GPO - Network Drives) already maps a department drive letter by OU. Update those drive mappings to point at the correct department subfolder so the mapped drive and the NTFS permission always match:

```
IT      → C: → \\FILESRV01\Departments\IT      (GG_IT)
HR      → C: → \\FILESRV01\Departments\HR      (GG_HR)
Management → C: → \\FILESRV01\Departments\Management (GG_Management)
```

Note: If a user's mapped drive shows Access Denied, check the group in Section 38.4 before changing any NTFS permission. Most drive-mapping problems are a missing group membership, not a broken permission.

38.10 Verify Permissions

From an IT workstation, confirm group membership:

```
Get-ADGroupMember -Identity GG_HR
```

Log on as a test user and confirm access:

```
Alice (GG_HR)      → can open  \\FILESRV01\Departments\HR
Alice (GG_HR)      → cannot open \\FILESRV01\Departments\IT
Vince (GG_IT)      → can open  \\FILESRV01\Departments\IT
Any domain user    → can open  \\FILESRV01\CompanyData\Public
```

If access does not behave as expected, run:

```
gpupdate /force
gpresult /r
```

on the client, and confirm the user's group membership has replicated to the domain controller the client is authenticating against.

39. GPO - Print Server Security

Target:

Servers

- Print Servers

For PRINT01:

Firewall → ON
Inbound → Block by default
Outbound → Allow

Windows Update → enabled/not blocked
Defender → enabled
Auditing → enabled

Note: Do not apply the workstation USB restriction to the print server unless specifically required.

40. GPO - Domain Controller Security

Target:

Domain Controllers

For:

DC01
DC02

Keep the first security baseline conservative.

Configure:

Windows Defender
→ Enabled/default protection

Windows Firewall
→ Enabled

Security Auditing
→ Success + Failure where appropriate

Avoid changing advanced authentication settings until their impact is understood.

Be especially careful with:

- User Rights Assignment
- Kerberos policies
- LDAP security
- NTLM settings
- SMB security
- Authentication protocols

Note: Incorrect configuration can disrupt domain authentication.

41. Domain Password Policy

Password policy is a special case.

Note: Do not create a GPO linked only to Company Users and expect it to control normal domain password requirements.

For the basic lab:

Group Policy Management

- lab.local
- Default Domain Policy
- Edit

Go to:

Computer Configuration

- Policies
- Windows Settings
- Security Settings

→ Account Policies
→ Password Policy

Example lab values:

Minimum password length
→ 8

Password must meet complexity requirements
→ Enabled

Enforce password history
→ 5

Maximum password age
→ 90 days

For a production environment, password requirements should follow the organization's approved security standard rather than blindly copying lab values.

42. Account Lockout Policy

Go to:

Account Policies
→ Account Lockout Policy

Example lab configuration:

Account lockout threshold
→ 5 invalid attempts

Account lockout duration
→ 15 minutes

Reset account lockout counter after
→ 15 minutes

Test:

```
1 incorrect password
2 incorrect password
3 incorrect password
4 incorrect password
5 incorrect password
      ↓
Account locked
```

This is useful for learning AD authentication and troubleshooting.

43. Applying and Troubleshooting GPOs

After changing a GPO, run on the affected computer:

```
gpupdate /force
```

Then:

```
gpresult /r
```

For a detailed HTML report:

```
gpresult /h C:\gpresult.html
```

Open:

```
C:\gpresult.html
```

Review:

- Applied GPOs
- Denied GPOs
- Computer policies
- User policies
- Security filtering
- OU targeting

44. Basic GPO Troubleshooting Workflow

When a user reports that a policy is not working:

1. Confirm the computer/user is in the correct OU
- ↓
2. Confirm the GPO is linked to that OU
- ↓
3. Confirm the GPO is enabled
- ↓
4. Run gpupdate /force
- ↓
5. Run gpresult /r
- ↓
6. Check security filtering
- ↓
7. Check whether another GPO overrides the setting
- ↓
8. Check Event Viewer if necessary
- ↓
9. Test the actual policy behavior

Note: Do not immediately change random settings. Identify which policy is responsible first.

45. Basic DNS Troubleshooting Workflow

If a computer cannot join the domain:

1. Check IP address
- ↓
2. Check subnet mask
- ↓
3. Check gateway
- ↓
4. Check DNS server
- ↓
5. Test DC01 by IP
- ↓
6. Test dc01.lab.local
- ↓
7. Test lab.local
- ↓
8. Test AD SRV records
- ↓
9. Attempt domain join again

Useful commands:

```
ipconfig /all
ping 192.168.1.100
nslookup lab.local
nslookup dc01.lab.local
```

For AD-specific DNS discovery:

```
nslookup
set type=SRV
_ldap._tcp.dc._msdcs.lab.local
```

46. Common Problems and First Checks

Problem	First thing to check
Client cannot join domain	Client DNS points to DC01
lab.local does not resolve	DNS service and DNS records
dc01.lab.local does not resolve	DC DNS record
GPO does not apply	OU placement and gpresult /r
RDP does not work	RDP GPO + firewall rule + user membership
User cannot RDP	GG-RemoteDesktop-Users membership
USB is blocked unexpectedly	GPO - Removable Storage
Wallpaper does not appear	File path and permissions

Problem	First thing to check
Network drive does not appear	Share path, permissions, and GPO application
Account is locked	Account lockout policy and authentication attempts

47. Security Principles

The following principles should be followed in a real organization:

Least Privilege

Users should receive only the permissions required for their jobs.

Separate Administrative Accounts

Do not use Domain Administrator accounts for normal daily work.

Group-Based Access

Prefer security groups instead of assigning permissions individually whenever practical.

Example:

GG-RemoteDesktop-Users

Centralized Configuration

Use GPOs for settings that need to be consistently applied across managed computers.

Change Carefully

Changes to:

- Kerberos
- LDAP
- NTLM
- SMB
- User Rights Assignment

can affect authentication and should be tested before production deployment.

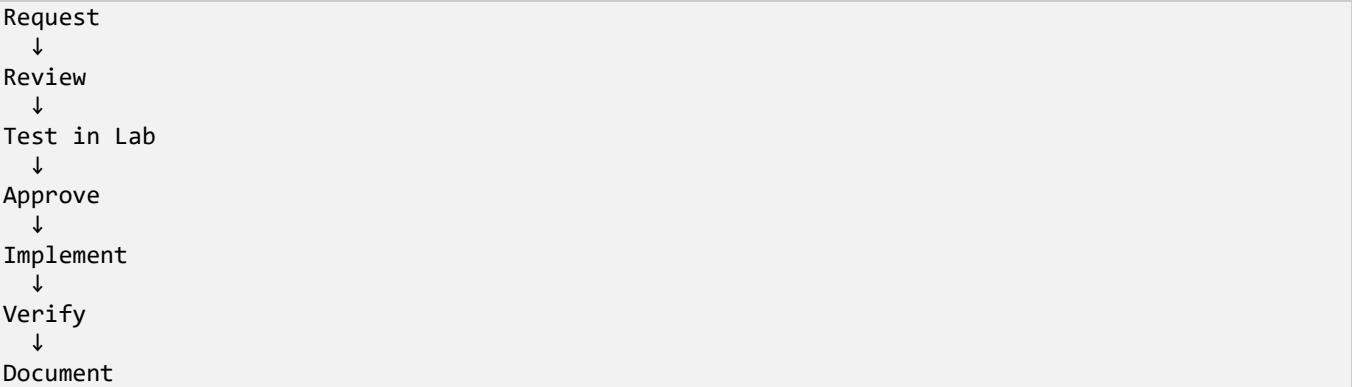
Document Changes

Record:

- What was changed
- Why it was changed
- Which GPO/server was affected
- Who performed the change
- How it was tested
- How it can be rolled back

48. IT Change Procedure

For significant AD/DNS/GPO changes:



Avoid making untested security changes directly on the production domain controller.

49. End-User Quick Guide

How do I log in?

Use your company domain account:

LAB\username

or:

username@lab.local

Why can't I access Control Panel?

Your computer may be receiving a company security policy.

Contact IT if you believe you need access.

Why can't I use a USB device?

Your workstation may have a removable-storage security policy.

Contact IT if you require approved removable media access.

Why is my computer asking for a password after inactivity?

The organization may require a screen lock after a defined period.

Why can't I use Remote Desktop?

RDP access is controlled by IT. Your account must be authorized.

Who should I contact?

Report the issue to the organization's IT Help Desk or assigned IT Support team.

Provide:

- Your name
- Computer name
- Problem description
- When the problem started
- Error message
- What you were doing when it occurred

50. Recommended Initial Implementation Order

Do not configure everything at once.

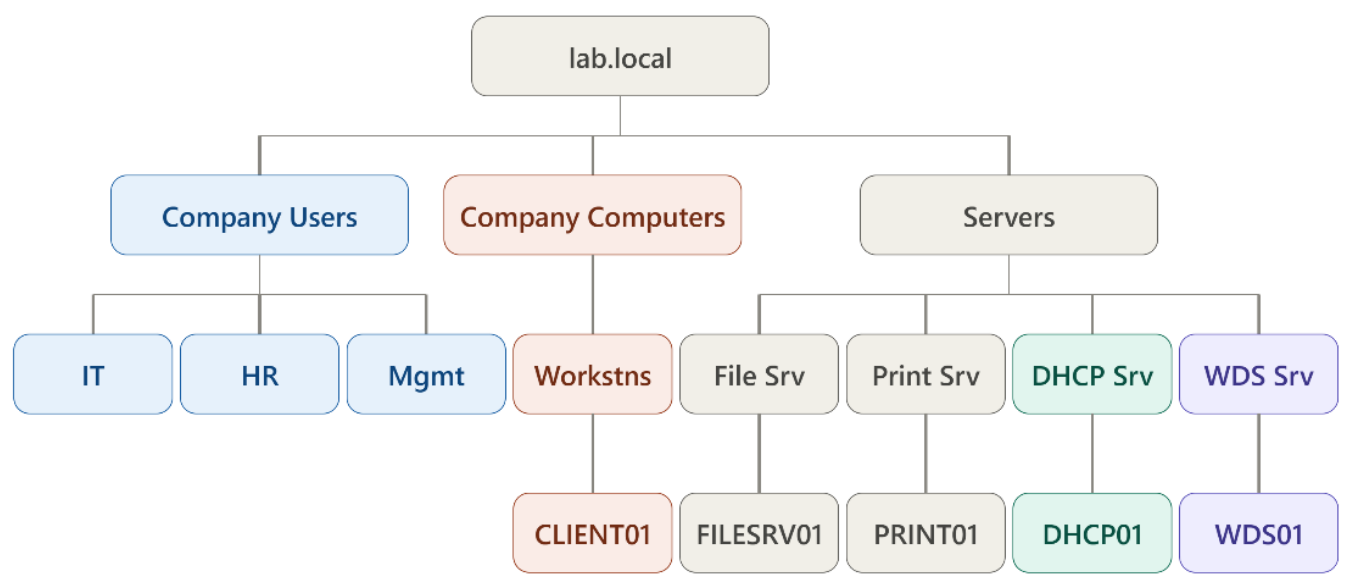
Use this order:

1. Configure DC01 static IP
↓
2. Rename DC01
↓
3. Install AD DS
↓
4. Promote DC01
↓
5. Verify DNS
↓
6. Configure client DNS
↓
7. Join CLIENT01 to the domain
↓
8. Create OUs
↓
9. Create security groups
↓
10. Create basic GPOs
↓
11. Configure RDP GPO
↓
12. Apply and test GPOs
↓
13. Configure server GPOs
↓
14. Configure domain password/lockout policy
↓
15. Configure conservative DC security

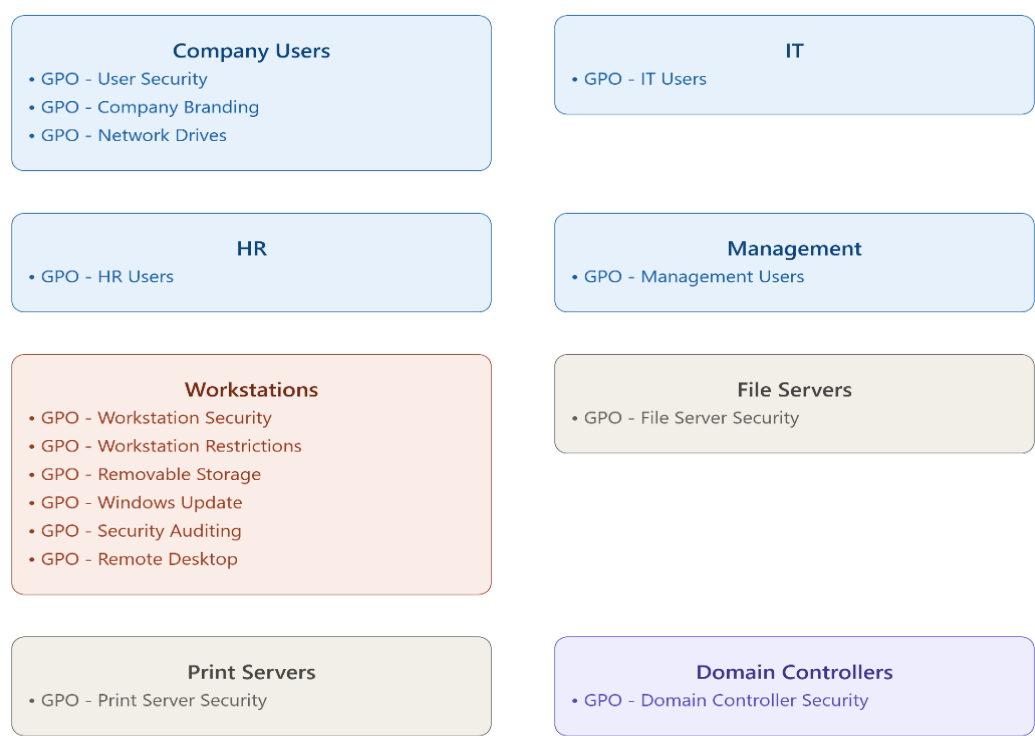
↓

16. Document and test

51. Final Configuration Map



GPO structure:



52. Initial Testing Checklist

AD DS / DNS

- ☐ DC01 has a static IP
- ☐ DC01 hostname is correct
- ☐ AD DS is installed
- ☐ lab.local exists
- ☐ DNS service is running
- ☐ lab.local resolves
- ☐ dc01.lab.local resolves
- ☐ AD SRV records resolve
- ☐ CLIENT01 uses 192.168.1.100 for DNS
- ☐ CLIENT01 successfully joined lab.local

Active Directory

- ☐ Company Users OU exists
- ☐ IT OU exists
- ☐ HR OU exists
- ☐ Management OU exists
- ☐ Company Computers OU exists
- ☐ Workstations OU exists
- ☐ Servers OU exists
- ☐ File Server OU exists
- ☐ Print Server OU exists
- ☐ DHCP Server OU exists
- ☐ WDS Server OU exists
- ☐ Required security groups exist
- ☐ Example users created in each OU (Vince, Admin, Alice, Bob, Manager)
- ☐ New users verified with Get-ADUser
- ☐ Test user added to GG-RemoteDesktop-Users where required

GPO

- ☐ Workstation Security created
- ☐ Workstation Restrictions created
- ☐ User Security created
- ☐ Company Branding created
- ☐ Network Drives created
- ☐ Removable Storage created
- ☐ Windows Update created
- ☐ Security Auditing created
- ☐ IT Users created
- ☐ HR Users created
- ☐ Management Users created
- ☐ Remote Desktop created
- ☐ File Server Security created
- ☐ Print Server Security created
- ☐ Domain Controller Security created

File Shares (FILESRV01)

- ☐ GG_IT, GG_HR, GG_Finance, GG_Management groups created
- ☐ Users added to their matching department group
- ☐ CompanyData folder structure created on FILESRV01
- ☐ NTFS permissions applied per Section 38.6
- ☐ CompanyData share created and shared with Authenticated Users
- ☐ Department drive mappings point to the correct department folder
- ☐ Test user can access their own department folder
- ☐ Test user cannot access another department's folder
- ☐ Every domain user can access Public

RDP

- ☐ RDP GPO linked to Workstations
- ☐ RDP enabled
- ☐ NLA enabled
- ☐ RDP firewall rule enabled
- ☐ GG-RemoteDesktop-Users created

- ☐ Authorized user added to the group
- ☐ Local Remote Desktop Users group contains the AD group
- ☐ gpupdate /force completed
- ☐ gpresult /r confirms the RDP GPO
- ☐ RDP connection tested

Security

- ☐ Firewall enabled
- ☐ Security auditing configured
- ☐ Password policy configured
- ☐ Account lockout policy configured
- ☐ Administrative privileges are limited
- ☐ Advanced authentication settings were not changed without testing

53. Most Important Beginner Exercises

If this is a learning lab, start with these:

1. Build DC01 with AD DS and DNS.
2. Create lab.local.
3. Configure CLIENT01 to use 192.168.1.100 as DNS.
4. Join CLIENT01 to the domain.
5. Create the OU structure.
6. Create a domain user.
7. Create GG-RemoteDesktop-Users.
8. Enable RDP through GPO - Remote Desktop.
9. Apply a workstation firewall policy.
10. Apply the 10-minute screen-lock policy.
11. Apply a company wallpaper.
12. Map a network drive.
13. Create the GG_IT, GG_HR, and GG_Management groups.
14. Apply NTFS permissions to each department folder on FILESRV01.
15. Test USB restrictions.
16. Test security auditing.
17. Use gpresult /r to troubleshoot policy application.
18. Test account lockout and recovery.

These exercises cover the core concepts of:

Active Directory	
	+
DNS	
	+
OUs	
	+
Security Groups	
	+
GPO	
	+
RDP	
	+
Windows Security	
	+
Troubleshooting	

54. Quick Reference Commands

Network

```
ipconfig /all
ping 192.168.1.100
nslookup lab.local
nslookup dc01.lab.local
```

Group Policy

```
gpupdate /force
gpresult /r
gpresult /h C:\gpresult.html
```

Domain Information

```
Get-ADDomain
```

AD DS Service

```
Get-Service NTDS
```

DNS Service

```
Get-Service DNS
```

RDP Status

```
Get-ItemProperty "HKLM:\System\CurrentControlSet\Control\Terminal Server" -Name fDenyTSConnections
```

RDP Group

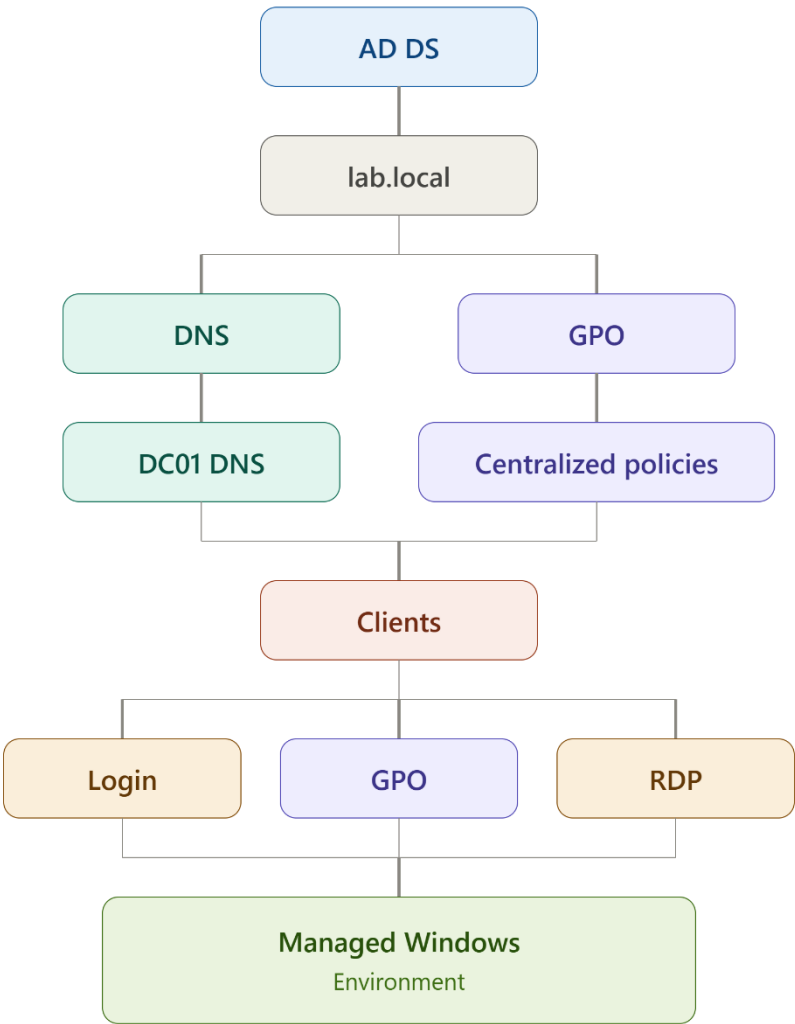
```
net localgroup "Remote Desktop Users"
```

Restart

```
shutdown /r /t 0
```

55. Summary

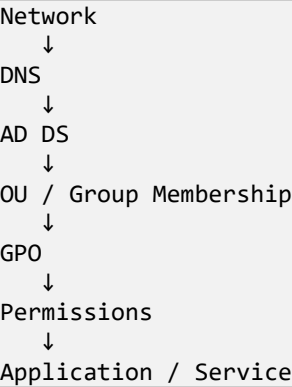
The complete environment follows this relationship:



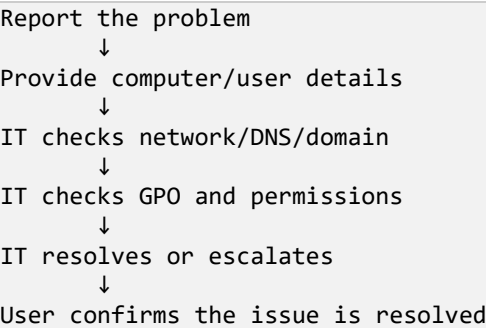
The key principle is:

Key Principle: AD DS provides identity and centralized management, DNS allows domain computers to locate AD services, OUs organize users and computers, and GPOs centrally apply configuration and security policies.

For an IT administrator, the primary troubleshooting sequence should be:



For a non-IT user, the important process is simpler:



This provides a practical foundation for expanding the lab into additional services such as DHCP, WDS, file services, print services, VPN, and centralized endpoint management.